

STA 332/MATH 343: Homework 2 – Solutions

Duke University, Fall 2026

Due: September 15th, 11:59 PM on Gradescope

Probability and Data Reduction

1. Let $X \sim \text{Bin}(m, p)$. Starting from the probability mass function, compute $\mathbb{E}[X]$ and $\text{Var}(X)$.

By the definition of expectation,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k=0}^m k \binom{m}{k} p^k (1-p)^{m-k} \\ &= mp \sum_{k=1}^m \binom{m-1}{k-1} p^{k-1} (1-p)^{m-k} = mp.\end{aligned}$$

Also, using $k^2 = k(k-1) + k$,

$$\begin{aligned}\mathbb{E}[X^2] &= \sum_{k=0}^m k(k-1) \binom{m}{k} p^k (1-p)^{m-k} + \mathbb{E}[X] \\ &= m(m-1)p^2 \sum_{k=2}^m \binom{m-2}{k-2} p^{k-2} (1-p)^{m-k} + mp \\ &= m(m-1)p^2 + mp.\end{aligned}$$

Therefore,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = m(m-1)p^2 + mp - m^2p^2 = mp(1-p).$$

2. Let $n \geq 4$, and suppose

$$T(\underline{X}) = (T_1(\underline{X}), T_2(\underline{X}))$$

is sufficient for a parameter θ . Determine which of the following statistics must be sufficient for θ . Justify each answer.

$$S_1(\underline{X}) = (T_1(\underline{X}), e^{T_2(\underline{X})}), \quad S_2(\underline{X}) = (-T_1(\underline{X}), T_2(\underline{X}), X_4), \quad S_3(\underline{X}) = e^{T_2(\underline{X})}.$$

S_1 must be sufficient because

$$T(\underline{X}) = ((S_1)_1, \log((S_1)_2)),$$

so T is a function of S_1 . Similarly, S_2 must be sufficient because

$$T(\underline{X}) = (-(S_2)_1, (S_2)_2).$$

The extra coordinate X_4 does not destroy sufficiency.

S_3 need not be sufficient: in general it discards T_1 . For example, if T_1 contains information about θ that is not determined by T_2 , the conditional distribution of the sample given S_3 can still depend on θ . Thus S_3 is guaranteed to be sufficient only under an additional condition ensuring that T_1 is a function of T_2 .

3. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\theta)$, where $\theta > 0$, and let

$$T = \sum_{i=1}^n X_i.$$

- (a) Find the distribution of T . *Hint: use the result of problem 13(a) on HW1.*
 (b) Without using the factorization theorem, prove directly from the conditional definition that T is sufficient for θ .

- (a) Sums of independent Poisson random variables are Poisson, so

$$T \sim \text{Poisson}(n\theta).$$

- (b) Fix $t \in \{0, 1, 2, \dots\}$ and a vector $\underline{x}$ of nonnegative integers. If $\sum_i x_i \neq t$, the conditional probability is zero. If $\sum_i x_i = t$, then

$$\begin{aligned} \mathbb{P}_\theta(\underline{X} = \underline{x} \mid T = t) &= \frac{\prod_{i=1}^n e^{-\theta} \theta^{x_i} / x_i!}{e^{-n\theta} (n\theta)^t / t!} \\ &= \frac{t!}{x_1! \cdots x_n!} \left(\frac{1}{n}\right)^t. \end{aligned}$$

This conditional probability does not depend on θ . Therefore, the conditional distribution of $\underline{X}$ given T is parameter-free, and T is sufficient for θ .

Exponential Families, Ancillarity, and Minimal Sufficiency

4. Let $Z_1, \dots, Z_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \tau)$, where $\tau > 0$ is the variance and $n \geq 2$, and suppose that we observe

$$X_i = Z_i^2, \quad i = 1, \dots, n.$$

- (a) Show that $X_i \sim \text{Gamma}(\frac{1}{2}, \frac{1}{2\tau})$.
 (b) Let f_τ be the pdf of the X_i . Show that $\{f_\tau(x) : \tau > 0\}$ is a scale family with scale parameter τ .
 (c) Use the likelihood-ratio criterion to prove that $T = \sum_{i=1}^n X_i$ is minimal sufficient for τ .
 (d) Show that

$$A = \frac{X_1}{\sum_{i=1}^n X_i}$$

is ancillary for τ .

- (a) For $x > 0$, the transformation $x = z^2$ has the two inverse branches $z = \sqrt{x}$ and $z = -\sqrt{x}$. Therefore,

$$\begin{aligned} f_{X_i, \tau}(x) &= f_{Z_i, \tau}(\sqrt{x}) \frac{1}{2\sqrt{x}} + f_{Z_i, \tau}(-\sqrt{x}) \frac{1}{2\sqrt{x}} \\ &= \frac{1}{\sqrt{2\pi\tau x}} \exp\left\{-\frac{x}{2\tau}\right\} \mathbb{1}_{\{x>0\}}. \end{aligned}$$

This is the density of

$$X_i \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2\tau}\right),$$

where the second parameter is the rate.

- (b) Write $Z_i = \sqrt{\tau}V_i$, where $V_i \sim \mathcal{N}(0, 1)$. Then

$$X_i = Z_i^2 = \tau V_i^2.$$

The distribution of V_i^2 is fixed, so the distributions of X_i form a scale family with scale parameter τ . Although $\sqrt{\tau}$ scales Z_i , squaring the observation also squares that factor, so τ scales X_i .

- (c) For sample points $\underline{x}, \underline{y} \in (0, \infty)^n$, the likelihood ratio is

$$\frac{L_{\underline{x}}(\tau)}{L_{\underline{y}}(\tau)} = \left(\frac{\prod_{i=1}^n y_i}{\prod_{i=1}^n x_i} \right)^{1/2} \exp \left\{ -\frac{1}{2\tau} \left(\sum_{i=1}^n x_i - \sum_{i=1}^n y_i \right) \right\}.$$

The first factor is parameter-free. The exponential factor is independent of τ if and only if

$$\sum_{i=1}^n x_i = \sum_{i=1}^n y_i.$$

Therefore, the likelihood-ratio criterion shows that

$$T = \sum_{i=1}^n X_i = \sum_{i=1}^n Z_i^2$$

is minimal sufficient for τ .

- (d) Since the X_i are independent Gamma random variables with a common rate,

$$T \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2\tau}\right).$$

Equivalently, $T/\tau \sim \chi_n^2$.

- (e) From part (b), $X_i = \tau V_i^2$, where $V_1, \dots, V_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Therefore,

$$A = \frac{\tau V_1^2}{\sum_{i=1}^n \tau V_i^2} = \frac{V_1^2}{\sum_{i=1}^n V_i^2}.$$

The distribution of the expression on the right does not depend on τ , so A is ancillary for τ .

5. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \beta)$, where $\alpha > 0$ is known and $\beta > 0$ is the unknown rate. Thus,

$$f_{\beta}(x) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \mathbb{1}_{\{x>0\}}.$$

- (a) Write f_{β} in the form

$$f_{\beta}(x) = g(x) \exp\{b(\beta)R(x) - c(\beta)\},$$

and identify g, R, b , and c .

- (b) Identify the minimal sufficient statistic $R_n(\underline{x})$ and explain why it is minimal sufficient for β .
- (c) Use the exponential-family moment identity to find $\mathbb{E}_{\beta}[X_1]$.

- (a) We can write

$$f_{\beta}(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)} \mathbb{1}_{\{x>0\}} \exp\{(-\beta)x + \alpha \log \beta\}.$$

Thus one valid representation is

$$g(x) = \frac{x^{\alpha-1}}{\Gamma(\alpha)} \mathbb{1}_{\{x>0\}}, \quad R(x) = x, \quad b(\beta) = -\beta, \quad c(\beta) = -\alpha \log \beta.$$

- (b) Closure under random sampling gives the natural statistic

$$S_n = \sum_{i=1}^n X_i.$$

It is sufficient by the factorization theorem. The support is fixed, $g(x) > 0$ on $(0, \infty)$, and $b(\beta) = -\beta$ is not constant, so the minimal-sufficiency theorem for natural statistics shows that S_n is minimal sufficient.

- (c) Since
- $b'(\beta) = -1$
- and
- $c'(\beta) = -\alpha/\beta$
- ,

$$\mathbb{E}_{\beta}[X_1] = \frac{c'(\beta)}{b'(\beta)} = \frac{\alpha}{\beta}.$$

- (d) The sum of independent Gamma random variables with a common rate is Gamma, so

$$S_n \sim \text{Gamma}(n\alpha, \beta).$$

6. Let
- $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\theta, \theta + 1)$
- , where
- $\theta \in \mathbb{R}$
- .

- (a) Show that this is a location family.

- (b) Show that the sample range

$$A = X_{(n)} - X_{(1)}$$

is ancillary for θ .

- (c) Find a sufficient statistic for
- θ
- whose dimension does not depend on
- n
- .

- (d) Prove that

$$T(\underline{X}) = (X_{(1)}, X_{(n)})$$

is minimal sufficient for θ . Because likelihoods may be zero, compare the likelihood functions directly rather than dividing them.

- (a) If
- $Z \sim \text{Unif}(0, 1)$
- , then
- $X = \theta + Z \sim \text{Unif}(\theta, \theta + 1)$
- . Therefore,
- θ
- is a location parameter.

- (b) Write
- $X_i = \theta + Z_i$
- , where
- $Z_1, \dots, Z_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, 1)$
- . Then

$$A = (\theta + Z_{(n)}) - (\theta + Z_{(1)}) = Z_{(n)} - Z_{(1)}.$$

Its distribution does not depend on θ , so A is ancillary for θ .

- (c) The joint likelihood is

$$\begin{aligned} L_{\underline{x}}(\theta) &= \prod_{i=1}^n \mathbb{1}_{\{\theta \leq x_i \leq \theta+1\}} \\ &= \mathbb{1}_{\{x_{(n)}-1 \leq \theta \leq x_{(1)}\}}. \end{aligned}$$

It depends on the data only through $(x_{(1)}, x_{(n)})$, so $T = (X_{(1)}, X_{(n)})$ is sufficient by factorization.

- (d) Restrict attention to sample points that have positive likelihood for at least one value of θ . For such a point $\underline{x}$, $L_{\underline{x}}$ is the indicator of the nonempty interval

$$I_{\underline{x}} = [x_{(n)} - 1, x_{(1)}].$$

Two such likelihood functions are proportional if and only if their indicator intervals are identical. This happens if and only if

$$x_{(n)} - 1 = y_{(n)} - 1 \quad \text{and} \quad x_{(1)} = y_{(1)},$$

or equivalently, $T(\underline{x}) = T(\underline{y})$. The likelihood criterion therefore shows that T is minimal sufficient.

7. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\theta)$, where $\theta > 0$ and $n \geq 2$. Define

$$S(\underline{X}) = \sum_{i=1}^n X_i \quad \text{and} \quad V(\underline{X}) = \left(\sum_{i=1}^n X_i, X_1 \right).$$

- (a) Show that V is sufficient for θ .
 - (b) Show that V is not minimal sufficient by using the definition of minimal sufficiency and the fact that S is sufficient.
 - (c) Confirm the conclusion in part (b) using the likelihood-ratio criterion.
- (a) We can recover S from V by taking the first coordinate. Since S is sufficient for θ , any statistic from which S can be recovered is also sufficient. Therefore, V is sufficient.
 - (b) If V were minimal sufficient, it would have to be a function of every sufficient statistic, including S . Consider the two sample points

$$\underline{x} = (1, 0, 0, \dots, 0) \quad \text{and} \quad \underline{y} = (0, 1, 0, \dots, 0).$$

They satisfy $S(\underline{x}) = S(\underline{y}) = 1$, but

$$V(\underline{x}) = (1, 1) \neq (1, 0) = V(\underline{y}).$$

Thus V cannot be a function of S . Since S is sufficient, V is not minimal sufficient.

- (c) The likelihoods at these two sample points are identical:

$$L_{\underline{x}}(\theta) = e^{-n\theta}\theta = L_{\underline{y}}(\theta).$$

Hence the likelihoods are proportional even though $V(\underline{x}) \neq V(\underline{y})$. A minimal sufficient statistic must assign the same value to precisely those sample points having proportional likelihoods. Therefore, the likelihood-ratio criterion also shows that V is not minimal sufficient.